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Wave
propagation

Altermagnets

Non-
hermiticity

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On wave propagation in solids and altermagnets

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University of Amsterdam

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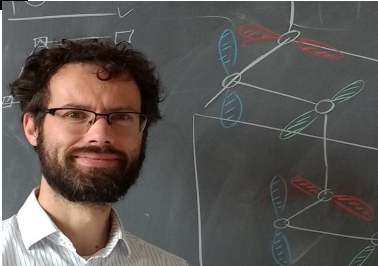
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Guide wave signals in solids



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Guide wave signals in solids

One can use different excitations: electrons, photons, phonons, etc.



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One can use different excitations: electrons, photons, phonons, etc.

1 Unidirectional



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One can use different excitations: electrons, photons, phonons, etc.

- 1 Unidirectional
- 2 Robust



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One can use different excitations: electrons, photons, phonons, etc.

- 1 Unidirectional
- 2 Robust $\longrightarrow$ topologically protected



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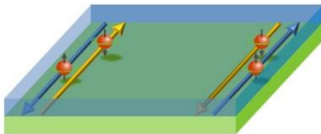
One can use different excitations: electrons, photons, phonons, etc.

- 1 Unidirectional
- 2 Robust $\longrightarrow$ topologically protected
- 3 Spin-momentum locking



Examples

Example 1: quantum spin Hall effect



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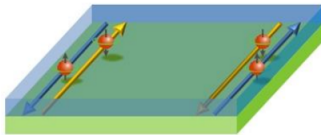
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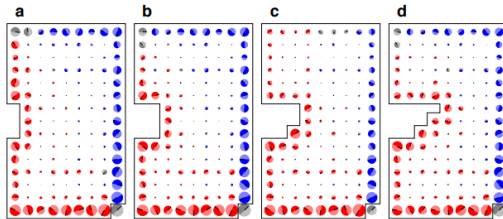


Examples

Example 1: quantum spin Hall effect



Example 2: 2D array of pendula



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What are altermagnets? A different phase of matter I

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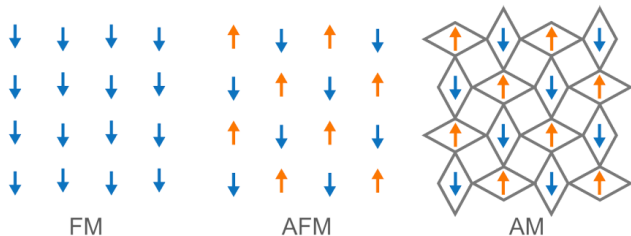
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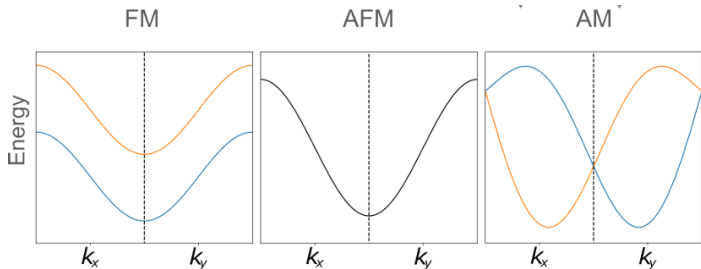


Kramer's theorem: $\text{TRS} + \text{half-integer spin} \implies \text{degenerate energies}$



What are altermagnets? A different phase of matter II

- 1 Symmetry: Rotation+TR denoted by $C_4\mathcal{T}$
- 2 Zero magnetization
- 3 Spin-split bands
- 4 $C_4\mathcal{T} \implies N_{\uparrow} = N_{\downarrow}$





Altermagnetic Lieb lattice

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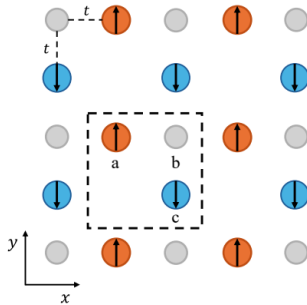
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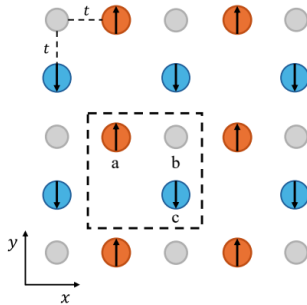
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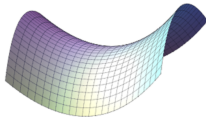
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It has hyperbolic bands:





Altermagnets have interesting properties

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1 Anisotropic wave propagation

- Unidirectional: control wave propagation direction
- Spin-momentum locking: different spins propagate in different directions

2 Piezomagnetism: magnetic moment is induced by strain

Useful in spintronics, spin-splitters, ultrafast photomagnetism, thermoelectrics ...



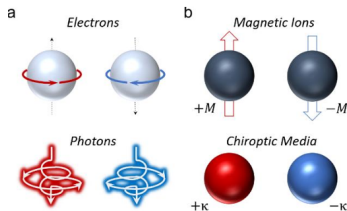
Generalization of alter symmetry

Generalization alter symmetry

spin $\rightarrow$ 2-state variable

Some examples are:

- Photon polarization $\rightarrow$ photonic altermagnet
- Electric dipole $\rightarrow$ alterelectric
- Pendula chirality $\rightarrow$ mechanical altermagnet (metamaterial)





Metamaterials I

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Example 1: piezoelectricity

- Onsager relations:

$$\varepsilon_{ij} = S_{ijkl}^E \sigma_{kl} + d_{ijk} E_k,$$

$$D_i = d_{ijk} \sigma_{jk} + \kappa_{ik}^\sigma E_k,$$

- Restrictions due to crystal symmetry:

$$[d_{ij}]^T = \begin{bmatrix} 0 & 0 & 0 & d_{15} & 0 \\ 0 & 0 & 0 & d_{24} & 0 \\ d_{31} & d_{32} & d_{33} & 0 & 0 \end{bmatrix}^T \quad (j = 1, 2, 3, 4, 5, 6)$$

- Enhancement due to complex structures
- Properties given by its structure, not by its parts



Metamaterials II

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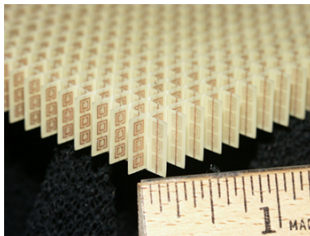
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Example 2: negative refractive index





Non-hermiticity

- Motivation: open systems

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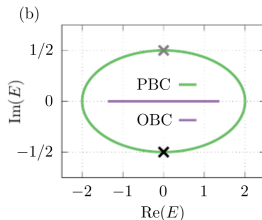
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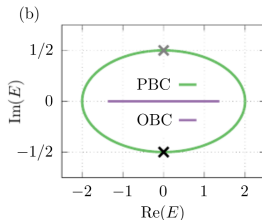
- Motivation: open systems
- Spectrum becomes complex





Non-hermiticity

- Motivation: open systems
- Spectrum becomes complex

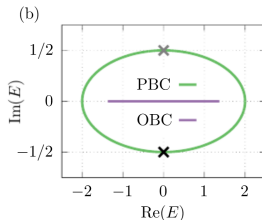


- **New** non-hermitian skin effect: eigenstates localize at boundary



Non-hermiticity

- Motivation: open systems
- Spectrum becomes complex

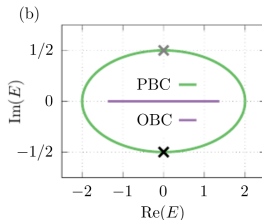


- **New** non-hermitian skin effect: eigenstates localize at boundary
- Example: Hatano-Nelson model



Non-hermiticity

- Motivation: open systems
- Spectrum becomes complex



- **New** non-hermitian skin effect: eigenstates localize at boundary
- Example: Hatano-Nelson model

$$\hat{H}_{\text{HN}} = \sum_j \left[t_L c_{j+1}^\dagger c_j + t_R c_j^\dagger c_{j+1} \right]$$



Non-reciprocity

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- Motivation: classical analogue of non-hermiticity



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- Motivation: classical analogue of non-hermiticity
- Breaks inversion symmetry $\rightarrow$ unidirectional waves



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- Breaks inversion symmetry $\rightarrow$ unidirectional waves
- Response differs when transmitter/receiver swapped



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- Motivation: classical analogue of non-hermiticity
- Breaks inversion symmetry $\rightarrow$ unidirectional waves
- Response differs when transmitter/receiver swapped
- Breaks Newton's third law (classical level)



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- $$\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} k & -k^o \\ +k^o & k \end{pmatrix} \begin{pmatrix} \Delta_1 \\ \Delta_2 \end{pmatrix}$$



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- Non-reciprocal units



Non-reciprocity

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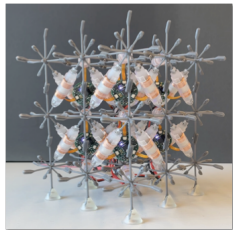
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 - $$\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} k & -k^o \\ +k^o & k \end{pmatrix} \begin{pmatrix} \Delta_1 \\ \Delta_2 \end{pmatrix}$$
- Non-reciprocal units
- Leads to pendula chirality





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The procedure I have been following is:

- 1 Choose a classical model



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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion



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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Multiple scale expansion



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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Multiple scale expansion
- 4 Map to quantum system



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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Multiple scale expansion
- 4 Map to quantum system
- 5 Analyze spectrum and eigenstates



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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
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- 4 Map to quantum system
- 5 Analyze spectrum and eigenstates

The idea is to apply them to increasingly more complex models.

Experimentally: build a Lieb lattice of non-reciprocal pendula and compare with theoretical predictions



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Goal

Study altermagnetism and non-hermiticity in classical analogues

- Altermagnets: $C_4\mathcal{T}$ symmetry can lead to unidirectional and spin-momentum locking propagation
- Non-hermitian skin effect: waves localized at boundaries
- Classical analogues enable experimental realization



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-  A. Visser, *Altermagnets and spinless analogs*, Master's thesis, University of Amsterdam (2025).
-  R. Süssstrunk and S. D. Huber, *Science* **349**, 47–50 (2015).
-  L. Smejkal, J. Sinova, and T. Jungwirth, *Physical Review X* **12**, 040501 (2022).
-  Y. Gao and Z. L. Wang, *Materials Today* **48**, 25 (2021).
-  M. Sato, Z. Gong, and K. Kawabata, *Physics Reports* **910**, 1 (2021).



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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.



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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.



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Activity: internal energy source.

There are different ways to achieve activity:



Activity

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Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping



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Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping
- Time delay



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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping
- Time delay
- **Non-reciprocity**



Non-reciprocal pendulum

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$$\ddot{x} + kx + \epsilon\gamma\dot{x} + \epsilon k^a y = 0 \quad (1)$$

$$\ddot{y} + ky + \epsilon\gamma\dot{y} - \epsilon k^a x = 0 \quad (2)$$

The energies are

$$\text{clockwise-die } E_1 = -\frac{i}{2} \left(\gamma + \frac{k^a}{\sqrt{k}} \right) \quad (3)$$

$$\text{anticlockwise-grow } E_2 = \frac{i}{2} \left(-\gamma + \frac{k^a}{\sqrt{k}} \right) \quad (4)$$

E_1 has 2 eigenvectors rotating clockwise with different initial phase- these modes die. E_2 has 2 eigenvectors that rotate anticlockwise with different initial phase- these modes grow.



2 interacting non-reciprocal pendula

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Out[58]//MatrixForm=

$$\begin{pmatrix} -\gamma & -\beta & 0 & \sigma & 0 & \beta & 0 & 0 \\ \beta & -\gamma & \sigma & 0 & -\beta & 0 & 0 & 0 \\ 0 & -\sigma & -\gamma & -r & 0 & 0 & 0 & r \\ -\sigma & 0 & r & -\gamma & 0 & 0 & -r & 0 \\ 0 & -\beta & 0 & 0 & -\gamma & \beta & 0 & \sigma \\ \beta & 0 & 0 & 0 & -\beta & -\gamma & \sigma & 0 \\ 0 & 0 & 0 & -r & 0 & -\sigma & -\gamma & r \\ 0 & 0 & r & 0 & -\sigma & 0 & -r & -\gamma \end{pmatrix}$$

In[59]:= **Eigenvalues**[M]

Out[59]= $\{-\gamma - i\sigma, -\gamma - i\sigma, -\gamma - i\sigma,$
 $-\gamma - i\sigma, -\gamma + i\sigma, -\gamma + i\sigma, -\gamma + i\sigma, -\gamma + i\sigma\}$

In[60]:= **Eigenvalues**[T2]

Out[60]= $\{-i\beta, i\beta, -ir, ir\}$



Model AM + NH I

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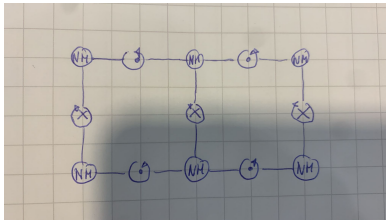
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We consider a Lieb lattice of non-reciprocal pendula.





Model AM + NH II

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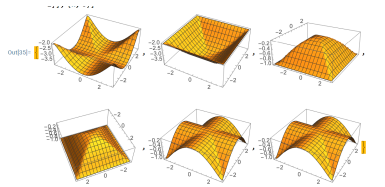
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It seems like there are modes with zero energy in certain directions- the band is flat; but not at $E=0$, I guess you can tune that with chemical potential

Other bands look more normal with somewhat well shape
Real spectrum:





Model AM + NH III

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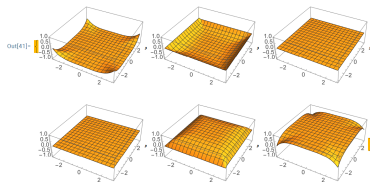
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Imaginary spectrum:





Mechanical altermagnet

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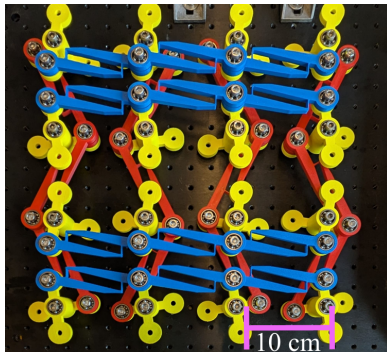
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- Focus on anisotropic wave propagation feature of altermagnets
- Exp: build on Amber's work: stack 2 toy layers
- Th: construct model and explain observations
- Study propagation of waves
- Main source: Amber's thesis





Piezoelectric

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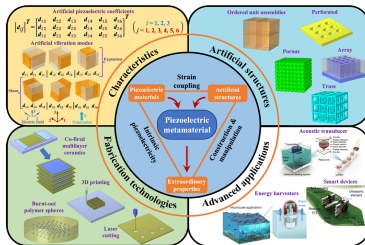
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- Exp: build altermaterial using capacitors with piezoelectric response
- Th: construct microscopic quantum model and compare
- Auxetic piezo
- Best case scenario: show piezoelectric properties of alterelectrics
- Review by Gao, Sharma





Non-hermiticity

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Annex

- Combine altermagnetic symmetry with non-hermiticity
- Exp: stack Yuan's cubes with odd elasticity
- Study waves propagating
- Gain-momentum locking
- Th: write analogous quantum hamiltonian and analyse it





Other

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Cervantes

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Dr. Corentin
Coulais

Wave
propagation

Altermagnets

Non-
hermiticity

Summary

Annex

- Metamaterials with higher order altermagnet symmetries
- Consider other rotations besides 90°
- For theory in general: band structure, ground state, excitations, topological properties, defects



Research plan

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Summary

Annex

- Calculate band structure of mechanical Lieb lattice
- Simulate Lieb lattice of non-reciprocal units
- Build Lieb lattice of non-reciprocal units

Possible questions

- Are there skin effects?



Do altermagnets always have hyperbolic dispersion?

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Annex

1 Break TRS



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- 1 Break TRS
- 2 C4T symmetry



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- 1 Break TRS
- 2 C4T symmetry
- 3 Inversion symmetry



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Summary

Annex

- 1 Break TRS
- 2 C4T symmetry
- 3 Inversion symmetry
- 4 Zero magnetisation